

2. 1

$$\text{pro } B = \left(\overset{(v_1)}{\begin{pmatrix} 1 \\ 2 \end{pmatrix}}, \overset{(v_2)}{\begin{pmatrix} 3 \\ 7 \end{pmatrix}} \right)$$

máme matici

$$X = \begin{pmatrix} 1 & 3 \\ 2 & 7 \end{pmatrix}$$

$$\begin{aligned} \langle x, y \rangle &= \left(X^{-1} x \right)^T \left(X^{-1} y \right) \\ &= \underline{x^T (X^{-T} X^{-1}) y} \end{aligned}$$

při ortogonalitě platí:

$$\rightarrow \langle x, y \rangle = x^T \begin{pmatrix} X^{-T} & X^{-1} \end{pmatrix} y$$

$$X = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$$

$$Y = \begin{pmatrix} y_1 \\ y_2 \end{pmatrix}$$

Ted vyřešme $X^{-T} \cdot X^{-1}$

$$X^{-1} = \begin{pmatrix} 7 & -3 \\ -2 & 1 \end{pmatrix}, X^{-T} = \begin{pmatrix} 7 & -2 \\ -3 & 1 \end{pmatrix}$$

$$\begin{aligned} X^{-T} X^{-1} &= \begin{pmatrix} 7 & -2 \\ -3 & 1 \end{pmatrix} \begin{pmatrix} 7 & -3 \\ -2 & 1 \end{pmatrix} = \\ &= \begin{pmatrix} 53 & -23 \\ -23 & 10 \end{pmatrix} \end{aligned}$$

$$\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}^T \begin{pmatrix} 53 & -23 \\ -23 & 10 \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} =$$

$$\begin{aligned} &= 53x_1y_1 - 23x_2x_1 - 23x_1x_2 \\ &\quad + 10x_2y_2 \end{aligned}$$